Name _____ Period _____ Date _____

2-3 Lesson Quiz: Factored Form of a Quadratic Function

short form Questions keep their Savvas quiz numbers. Show your work.

2. Solve the equation $x^2 + 7x = 30$.

Ⓐ $x = 15$ and $x = -2$

Ⓑ $x = -6$ and $x = 5$

Ⓒ $x = -10$ and $x = 3$

Ⓓ $x = 10$ and $x = -3$

Show the factored equation and the Zero Product Property step:

_____

_____

_____

3. A projectile is launched into the air. The function $h(t) = -16t^2 + 32t + 128$ gives the height, h, in feet, of the projectile t seconds after it is launched. After how many seconds will the projectile land back on the ground?

_____ seconds

Work:

4. Identify the intervals on which the function $f(x) = x^2 + 12x + 27$ is positive.

$(-\infty,$ _____) and (_____ , $\infty)$

Show the zeros and a test value in each interval:

Scoring: Question 2 and Question 4 are 10 points each. Question 3 is a bonus of up to 5 points (it uses the same function as Try It 4 in the packet). Total out of 20.

2-3 Lesson Quiz, short form: answer key

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2. C, $x = -10$ and $x = 3$. $x^2 + 7x - 30 = 0$, $(x + 10)(x - 3) = 0$, so $x = -10$ or $x = 3$. Distractor B solves $x^2 + x = 30$; D has both signs flipped, the error of reading the constants in $(x + 10)(x - 3)$ as the zeros. Savvas quiz item 2, DOK 2, Skills Review A24. Taught by Example 3 (cut); the packet rehearses get-0, factor and the Zero Product Property in Examples 4 and 5, Try Its 4 and 5, and the Exit question ($w(w + 5) = 50$). IXL CJC. **10 points:** 6 for C, 4 for the factored equation and the Zero Product Property step.

3. 4 seconds. $0 = -16t^2 + 32t + 128 = -16(t^2 - 2t - 8) = -16(t - 4)(t + 2)$, so $t = 4$ or $t = -2$; time cannot be negative, so the projectile lands after 4 seconds. Check: $-16(16) + 32(4) + 128 = -256 + 128 + 128 = 0$. Savvas quiz item 3, DOK 2, Skills Review A29. Example 4 and Try It 4 (the same function). IXL NMZ, CJC. **Bonus 5:** 2 for setting $h = 0$ and factoring out -16, 3 for 4 with the negative time rejected.

4. -9, -3; $(-\infty, -9)$ and $(-3, \infty)$. $x^2 + 12x + 27 = (x + 9)(x + 3)$, zeros -9 and -3. Test values: $x = -10$ gives $100 - 120 + 27 = 7 > 0$; $x = -5$ gives $25 - 60 + 27 = -8 < 0$; $x = 0$ gives $27 > 0$. Savvas quiz item 4, DOK 2, Skills Review A70. Example 5 and Try It 5. IXL DP9. **10 points:** 4 for the zeros -9 and -3 (the factored form), 2 for a test value in each interval, 4 for the two intervals.

Items left off this form (available for a retake or Form B):

- Savvas item 1, factor $3x^2 - 11x - 4$: answer 3, 1, 4, that is $(3x + 1)(x - 4)$ ($ac = -12$; 1 and -12 have sum -11), DOK 2, A24, A69; taught by Example 1 (cut). IXL UB5.
- Savvas item 5, the long-jumper parabola: answer A, $y = -x^2 + 12x - 27$ (zeros 3 and $3 + 6 = 9$, the point $(8, 5)$ gives $5 = -5a$, $a = -1$), DOK 2, F50, F58; taught by Example 6 (cut). IXL HPB.

IXL replacement: SmartScore 70 or above on NMZ, CJC and DP9 (the starred skills; NMZ feeds the item 3 bonus) replaces this quiz with 20/20.

Verify every key against the Savvas Realize quiz PDF (aga_24_a2_na_0203_lq.pdf) on quiz day; this form is built from the TE transcription (a2_2-3_TE.tex, printed page 82A).